Confirmation Bias and the Plausible Distribution of Evidence

Abstract

This paper discusses different, inter-related notions of confirmation bias as classified by Nickerson (1998), distinguishing broadly between confirmation bias in evidence search, evidence selection, evidence evaluation, and evidence communication. It then sets out an overlooked rational basis for seeming bias in evidence search that stems from reasonable expectations about the real-world distribution of evidence in favour and against a claim. The implications of this for communication and argumentation are discussed, as are the implications of taking into account plausible evidence distributions for research on bias more generally.

Keywords: confirmation bias, myside bias, rationality, Bayesian argumentation

> "If one were to attempt to identify a single problematic aspect of human reasoning that deserves attention above all others, the confirmation bias would have to be among the candidates for consideration."
>
> Nickerson, 1998 pg. 175

1 Introduction

As indicated in the opening quote, drawn from Nickerson's extensive review (Nickerson, 1998), few, if any, phenomena are more paradigmatically associated with human

bias and irrationality than confirmation bias. Unsurprisingly, given the centrality of confirmation bias, many overlapping aspects to it have emerged along with corresponding empirical paradigms probing them. On an abstract, conceptual, level these different aspects can be distinguished by considering the two distinct components involved in the formation of beliefs on the basis of evidence: evidence acquisition and evidence evaluation. Since most hypotheses of interest will involve multiple pieces of evidence, evidence evaluation entails selection and weighting of evidence in the process of aggregation. Likewise, the existence of multiple pieces of potential evidence that likely vary both in strength and polarity (for or against the hypothesis) means that evidence acquisition will involve complex decisions about search.

There are consequently multiple conceptually distinct processes that can be subject to confirmatory bias. By the same token, there are also distinct explanatory routes for alternative, rational, explanations of patterns that –at first glance– seem indicative of bias. This paper seeks to clarify a key underlying factor constraining evidence search and evidence acquisition, and, through that, constraining resultant beliefs. This factor is the expected distribution of evidence in the world.

Specifically, it is shown how the expected distribution of evidence already features in a number of prominent debates about confirmation bias. The main contribution of the paper is then to provide a general treatment of rational expectations about the distribution of evidence, as follows from a Bayesian perspective. This more general perspective, in turn, sheds new light on empirical phenomena that have been viewed as some of the most robust examples of unduly confirmatory behaviour to date.

The paper proceeds in three parts: the first provides a brief overview of the multifaceted phenomenon that is 'confirmation bias' and highlights the role of Bayesian counter-explanations in past debate. The second, and main, part provides a general treatment of rational expectations about the distribution of evidence. The final part then discusses the implications.

2 A Brief History of Confirmation Bias

Along with the prominent position of confirmation bias comes a fragmentation into a host of inter-related conceptual and empirical sub-types which have variously been cast as different aspects or versions of the same underlying concept. That underlying concept, according to Nickerson (1998), involves the "inappropriate bolstering of hypotheses or beliefs whose truth is in question" (pg. 175). At root, confirmation bias involves inappropriate selectivity in the acquisition and use of evidence. It thus involves 'bias' in the very general sense of being unduly partial to something, in this case a particular hypothesis (on the many, interrelated notions of bias in psychology, see e.g., Hahn & Harris, 2014). This partiality, though, seems intuitively likely to give rise to bias in a more technical sense of expected systematic deviation of an estimate from a true value. The fact that an estimator is biased need in no way rule out its adaptive value (given the trade-off between bias and variance a biased estimator may be more accurate on average, in the sense of closer to the true value, than an unbiased one, see Hahn & Harris, 2014); however, the kinds of specific manifestations of 'confirmation bias' that psychology has come to distinguish typically suggest violations of fundamental norms of rationality such as logic, probability theory, or decision theory (on norms of rationality see e..g., Hahn, 2026). Conversely, counter-evidence against seeming manifestations of confirmation bias may involve a demonstration that the pattern of behaviour in question can, in fact, be produced by an agent following these rationality norms.

Figure 1 shows a typology of confirmation bias that follows the section headings of Nickerson's seminal review, supplemented with a third column that holds counter-evidence, including counter-evidence that has been put forward since publication 25 years ago.

Taking simply the section headers Nickerson's comprehensive review used for extant empirical work at the time, there are 4 broad areas of research that might

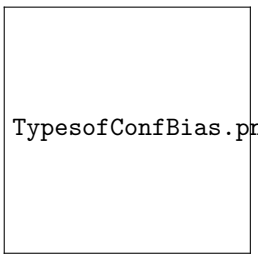


Fig. 1 Column 1 (blue) shows a typology of phenomena that have been taken to indicate confirmation bias based on the organisation of material in Nickerson's comprehensive (1998) review. The second column (green) provides a reference to a particular study showing this phenomenon. The third column (orange) lists sample studies providing counter-arguments against seeming irrationality.

be distinguished: 1. Hypothesis-determined information seeking and interpretation; 2. debate surrounding Wason's selection task and formal reasoning; 3. experimental evidence for primacy effects and belief persistence whereby initially formed opinions or beliefs seem unduly resistant to change, and, finally, work on 4. overconfidence and the illusion of validity.

These headers themselves provide a rough organising framework for further sub-literatures. In particular, the bulk of psychological research has focussed on what Nickerson summarises under 'hypothesis determined information seeking and testing', comprising seven different areas of substantial research. These areas are themselves only loosely connected. They not only cut across different parts of psychology (e.g., cognitive psychology, social psychology), even within those they fall into different specialisations (e.g., reasoning versus judgment and decision-making research). This, in and of itself, highlights how pervasive the concern with confirmation bias has been.

Arguably, each of the sub-areas highlighted by Nickerson are, 25 years later, still areas of active research that continue to generate new accounts. Column 3 contains a subset of papers presenting counter-evidence with respect to the target phenomenon that seeks to establish either empirically or conceptually that the observed pattern does not, in fact, represent bias. Many of these studies have themselves attracted sizeable volumes of subsequent, critical work. Discussing these subsequent strands

would require multiple reviews, and the remainder of the paper will pick out only a small subset.

At the same time, there have been additions and changes to Nickerson's typology. The most significant development has, arguably, been the extent to which interest in conformation bias (and, relatedly, motivated cognition) has moved from an individual level concern to a concern about collectives by virtue of the explosion of interest in polarisation. Nickerson almost foreshadowed this in as much as he notes in the opening sections of the paper that "one is led to wonder whether the bias, by itself, might account for a significant fraction of disputes, altercations, and misunderstandings that occur among individuals, groups, and nations" (pg. 175).

However, it is notable that, bar fewer than a handful of references, the material that features in Nickerson's extensive review concerns exclusively individual level phenomena, and the word polarisation appears only twice in the manuscript, within the reference section.

Since publication, there have been hundreds of papers on polarisation (for a review, see e.g., (Sunstein, 2009)). In that literature, confirmation bias (and, relatedly, motivated reasoning) are not just routinely invoked as causal drivers of polarisation, the mere fact of group polarisation is itself frequently taken as evidence of confirmation bias and motivated reasoning.

2.1 Bayesian Perspectives on Confirmation Bias

In considering the types of counter-arguments that have been produced against various forms of confirmation bias (Fig. 1, Column 3), it is noteworthy that many of those arguments have, in effect, been statistical. Specifically, they draw, in one way or another, on the tools of statistics and probability theory to demonstrate that what might, at first blush, seem biased or irrational, is, in fact, rational or adaptive under a particular set of assumptions. In other words, counter-argument has taken the form

of showing that an optimal agent, who is rational by definition, shows the same behavioural patterns. The typical formulation of such agents is Bayesian.

2.1.1 Bayesian Framework

At its core, 'being Bayesian' means assigning probabilities in line with the axioms of probability and drawing on Bayesian conditionalisation for belief revision, that is, updating beliefs in light of new evidence in accordance with Bayes' rule (Pettigrew, 2016).

Around that, a broader family of perspectives exist that vary with respect to whether they view probabilities as expressions of subjective degrees of belief or additionally or alternatively as in some way objective (Williamson, 2010) and that vary in the extent to which they countenance other rules for belief revision in addition (e.g., Jeffrey conditionalisation, (Jeffrey, 1990), or distance minimisation, see (Eva & Hartmann, 2018)).

Bayesian reconstructions of cognitive tasks now abound (Chater, Oaksford, Hahn, & Heit, 2010; Gopnik & Bonawitz, 2015). The most relevant for present purposes, however, is the application of the Bayesian framework to argument (Hahn, 2020). In essence, Bayesian argumentation treats arguments as evidence and uses the Bayesian framework to evaluate how strong those arguments are, that is, it asks how much those arguments *should* change our beliefs. This approach has been used to elucidate everything from logical fallacies (Chater & Oaksford, 2000), through well known fallacies such as arguments from ignorance (Hahn & Oaksford, 2008) or slippery slope arguments (Corner, Hahn, & Oaksford, 2011) or ad hominems (Bhatia & Oaksford, 2015), to plausible default schemes of everyday argument (for a review see, (Hahn, 2020)).

On Bayes' rule, the extent to which a piece of evidence should change our subjective degree of belief in a hypothesis is determined by how diagnostic the evidence is, that is, how much more likely it is if the hypothesis is true $P(e|H)$ than if it is false $P(e|\neg H)$:

$P(e\|H)$ True Positive	$P(\neg e\|H)$ Incorrect Rejection
$P(e\|\neg H)$ False Positive	$P(\neg e\|\neg H)$ Correct Rejection

Table 1 The four different types of evidence in the terminology of signal detection theory and their associated likelihoods.

$$P(H|e) = \frac{P(e|H) * P(H)}{P(e)} = \frac{P(e|H) * P(H)}{P(e|H) * P(H) + P(e|\neg H) * P(\neg H)}$$

Increasing $P(e|H)$ and decreasing $P(e|\neg H)$ will increase the posterior $P(H|e)$, so the likelihood ratio $P(e|H)/P(e|\neg H)$ provides a natural measure of the evidential value or strength. For example, H might be the presence of an underlying disease in a patient, and e a positive test result. The more likely such a positive test result is obtained *only* in genuine case of disease, the more diagnostic the test.

Terminologically, it will be useful in the remainder to refer to $P(e|H)$ as the sensitivity and $P(\neg e|\neg H)$ as the specificity in keeping with the terminology used to characterise medical tests, but also, additionally, to refer to all four conditional probabilities associated with a piece of evidence in signal detection terms (Green, Swets, et al., 1966) as set out in Table 1. That is, to continue with the medical example, a false positive is a positive test result obtained where the disease is not, in fact, present; conversely, an incorrect rejection is a negative test result obtained despite the patient having the disease. The probabilities of receiving such a false positive or an incorrect rejection are $P(e|\neg H)$ and $P(\neg e|H)$ respectively. Furthermore, unless otherwise indicated, 'evidence' and 'argument' will be used interchangeably. In as much as the distinction matters at all, one might think of evidence constituting the content of an argument, making evidence something we receive either 'from the world' or in the form of an argument, that is, through communication.

With the Bayesian framework in hand, one can distinguish two distinct explanatory roles assigned to it in the context of countering charges of confirmatory bias. The first of these roles pertains to evidence evaluation.

2.1.2 The Explanatory Role of Diverging Subjective Likelihoods

The one polarisation phenomenon Nickerson does discuss is so-called 'biased assimilation'. (Lord, Ross, & Lepper, 1979) provided experimental evidence that participants with opposing prior beliefs updated their beliefs in opposing directions given seemingly equivocal evidence that consisted of mixed, conflicting arguments. The intuitive interpretation of this finding assumed that participants selectively attend to, or give greater weight to, those arguments that confirm their prior position. In response, Jern, Chang, & Kemp, 2014, pointed out that rational Bayesians may draw different conclusions from the 'same' evidence given different underlying models of the world. In the same vein, (Hahn & Harris, 2014) pointed out that rational Bayesian agents who assign different likelihoods to the same evidence will diverge in the conclusions they draw. In as much as the sensitivity and specificity characterise the 'reliability' of a test, diverging views on source reliability will give rise to seemingly biased assimilation. Such divergence can readily occur in practice because the reliability of our sources may be difficult to ascertain. Among the suggested 'solutions' to this difficulty is that agents perform a joint inference over their belief in the truth or falsity of the hypothesis and the reliability of the source (for a review, see (Merdes, Von Sydow, & Hahn, 2021)). Such a strategy, in effect, links the perceived reliability of the source to the plausibility of their evidence in light of one's current degree of belief in the hypothesis. More generally, participants entering an experiment on 'biased assimilation' may not just have systematically different views on the diagnostic value or strength of the evidence depending on their prior opinion, they may, in fact, be rationally required to do so in order to be probabilistically coherent.

A variety of different models of group deliberation and communication across social networks have drawn on the possibility of divergent perceptions of source reliability to model not just how agents with diverging beliefs may polarise further given conflicting evidence, but, more importantly, how the same mechanism can give rise to

polarisation in initially homogenous populations merely as a function of agents' accidental history of encountering evidence (Hahn, Merdes, & von Sydow, 2024; Madsen, Bailey, & Pilditch, 2018; Pallavicini, Hallsson, & Kappel, 2021).

While these counter-positions on confirmation bias have focussed on evidence evaluation, other research has made use of the Bayesian framework in countering charges of confirmation bias, in particular in evidence search, by appealing to evidence distributions.

2.1.3 The Explanatory Role of Evidence Distributions

Different strands that might be summarised under this header involve counter-arguments to a charge of confirmation bias that appeal to objective constraints on priors and likelihoods themselves.

One of Nickerson's categories is 'looking only for positive instances' (see Fig. 1, 1.3). Such a tendency was famously demonstrated by (Wason, 1960) who found that participants required to infer an underlying rule from a sequences of instances preferred queries about instances that would affirm their currently favoured rule. The potential 'irrationality' here is that positive instances alone are insufficient to rule out overly general rules. This basic fact long shaped discussions of language acquisition and the innateness of a universal grammar because it meant that the 'correct grammar' underlying a language cannot be inferred solely through positive instances (grammatical sentences) (Gold, 1967). Explicit negative feedback (e.g., being informed that 'bark dog the' is not a grammatical sentence of English) to language learners is rare, giving rise to a 'poverty of stimulus' argument in favour of nativism. A probabilistic perspective that replaces correct identification of the 'true' underlying grammar with a high probability of identifying a grammar, however, fundamentally changes the picture, and the absence of labelled negative instances is no longer essential. Subsequent work has highlighted that there are likelihoods under which the persistent absence of an instance in the data (e.g., the absence of particular grammatical construction)

licenses the inference that it is not allowed under the rule(s) (Hahn & Oaksford, 2008; Hsu, Chater, & Vitányi, 2013; Perfors, Tenenbaum, & Wonnacott, 2010). Under such circumstances, at least, focussing on positive instances may be rational.

Wason subsequently pursued a similar theme with his 'selection task' (Wason, 1968) which became a focal point for reasoning research (see Fig. 1, column 1, 1.2). In the selection task, participants are asked to identify what information to seek in order to evaluate a conditional rule ('if a card shows a vowel on one side, there is an even number on the other side'). The logic of conditionals implies that only a falsificationist strategy can yield a definitive answer, yet, only a small minority of participants choose accordingly. (Oaksford & Chater, 1994), however, showed that participants' responses on the task reflect those of Bayesian optimal data selectors given appropriate assumptions about the conditional probabilities involving the truth or falsity of the rules antecedent ('if p') and consequent ('then q'), and the assumption that p and q are rare (the so-called 'rarity assumption').

Finally, seeming confirmation bias in information search in the context of judgments of contingency or correlation (e.g., Fig. 1, 1.1) has also been subjected to Bayesian counter-analysis (Crupi, Tentori, & Lombardi, 2009; Klayman & Ha, 1987) that rests on assumptions about likelihoods.

This breadth is unsurprising given that Klayman and Ha (1987) already showed that the same probabilistic framing can be applied to all three of these tasks. That probabilistic framing clarified also the difference between seeking to confirm a currently favoured belief or hypothesis, and seeking to test or examine instances that fall under that hypothesis ('positive test strategy') as opposed to ones that do not. As (Klayman & Ha, 1987) showed in very general terms, a positive test strategy does not intrinsically contradict the goal of falsification. In fact, there are circumstances where searching for positive instances may be the only route to falsification; and in probabilistic environments, it depends whether falsification provides more information

than verification. That 'depends' rests on priors and likelihoods and, with that, on the probability of the different kinds of possible evidence as set out in Table 1.

Appeals to the distribution of evidence as the basis for counter-arguments against confirmation bias are, however, not restricted to tasks involving search. Proper consideration of the distribution of evidence has also played a role in alternative explanations for seemingly biased judgments. Properties of participant judgments have been taken to be indicative of biased evaluation. We saw above the way in which diverging subjective likelihoods have been invoked to undercut the inference from polarised beliefs to irrationally biased evaluation. In parallel to this, appeals to the distribution of evidence encountered have served to undercut inference from patterns of beliefs to biased evaluation.

The first example of this in the context of confirmation bias concerns illusory correlations (see Figure 1 1.7). These are subjective estimates of correlations in excess of the objective correlation (Fiedler, 2000). They feature as examples of confirmation bias by virtue of the fact that a wealth of experimental evidence has suggested these subjective estimates to 'reflect a compromise between actual observations and prior expectancies' (see Fiedler, 2000, for a review). In this context, Fiedler (2000) presents a simple connectionist model that explains judgmental biases as arising from simple rules of associative learning in a probabilistic environment. This model can generate illusory correlations without biased processing, in the absence of any expectations. In particular, it can generate illusory correlations solely due to sample size and the extent to which particular event combinations are rare in its input experience.

The second example concerns optimism bias and optimistic belief updating. Fiedler lists optimism bias whereby individuals seemingly under-estimate their personal risk of undesirable, negative events (Weinstein, 1980) as an example of an illusory correlation. The standard explanation of unrealistic optimism appeals to motivated reasoning such that the processing of evidence is biased in order to reach a desired conclusion

(Weinstein, 1980). This, in turn, is one of the different explanations of confirmation bias ('desire to believe') that Nickerson highlights. (Harris & Hahn, 2011), however, show that the standard experimental method for ascertaining the presence of unrealistic optimism generates seemingly optimistic data patterns as a statistical artifact. Rational Bayesian agents will generate patterns of judgment indicative of optimism because of the expected distribution of evidence across individuals that suggests to them that they are at greater or lesser risk.

More recent work has sought to replace Weinstein's original method with more direct tests of biased processing by examining the impact on belief revision of participants receiving favourable or unfavourable information about event base rate (Sharot, Korn, & Dolan, 2011). These methods too, however, can give rise to wholly artefactual optimistic bias in rational agents because the implicit assumptions of experimental design and analysis about the prevalence of negative events and the distribution of diagnostic evidence concerning individual risk across a group of participants cannot mathematically be met ((Shah, Harris, Bird, Catmur, & Hahn, 2016); see also, for a related, but distinct failure (Lench & Ditto, 2008) and the discussion in (Hahn & Harris, 2014)). In particular, it is mathematically impossible to equate fully participants' diagnostic information concerning their own individual risk (e.g., a family history of a disease), participants' base rate error and the amount by which participants have to update their beliefs when informed of the correct base rate across participants over- as opposed to under-estimating the event base rate. Asymmetric belief revision, whereby Bayesian agents receiving 'good news' (a new lower base rate) as opposed to 'bad news' (a higher base rate than previously adopted) will update their beliefs by different amounts is consequently inevitable as long as those agents also possess diagnostic information about their own individual risk

In summary, two things can be concluded from the preceding overview. One is that the Bayesian framework has played a significant role in the continuing debate

about confirmation bias. This is to be expected as the baseline behaviour of a rational agent is the foil against which the irrationality of specific behaviours must be established. The second conclusion is that multiple distinct phenomena grouped under the header of 'confirmation bias' have seen counter-arguments to seeming bias and irrationality that involve, in one form or another, appeals to the distribution of evidence. Such arguments have been both non-Bayesian (e.g., Fiedler, 2000) and Bayesian (e.g., Oaksford & Chater, 1994). Typically, however, they have been *conditional*: given those distributional assumptions, the observed behaviour seems rational. Only some of the examples include attempts to justify that these assumptions are, in fact, likely to be met. An example of this is Oaksford & Chater's intuitive motivation of the rarity assumption, which itself prompted subsequent debate (Evans & Over, 1996; Laming, 1996; Oaksford & Chater, 1996). Or, in the case of unrealistic optimism research, the relevant constraints follow from a single empirical fact about the world, namely that there is an inverse correlation between just how bad an event is and its probability of occurrence (Harris & Hahn, 2011).

As the remainder of the paper seeks to establish, however, there are constraints on the expected distribution of evidence that follow directly from the Bayesian perspective on evidence or argument itself, that is, without regard to the properties of specific real-world content. We turn to this next.

3 Examining the Distribution of Evidence

The main argument of this paper is that the broader role of plausible evidence distributions has not yet received the full recognition it deserves. Not only is there little recognition of the common thread that unites many strands of counter-argument to seemingly irrationally confirmatory behaviour (as the preceding section has sought to draw out), the issue of constraints on evidence distributions has itself, arguably, not been explored in full depth. Rectifying this is the goal of the second, and main, part

of this paper. Specifically, it is the aim to show how the Bayesian framework draws out basic constraints on what kinds of evidence we could expect. These constraints, in turn, provide robust and significant constraints on possible patterns of rational behaviour that bias-focussed research must take into account.

3.1 Expected Evidence Distribution

The following exposition proceeds in two parts. First, it is shown how our expectations about the distribution of evidence or arguments relates to our current beliefs about the truth or falsity of the hypothesis at issue. Second, it is shown how thinking about the world as a generator of evidence constrains plausible evidence distributions. Both lines of argument highlight ways in which plausible evidence distributions, that is expectations about evidence, constrain rational behaviour. These are then drawn out in a third, and final, section.

3.1.1 Arguments in Favour of the True Hypothesis are more Likely than Arguments Against

The expected distribution of evidence is sensitive to our current beliefs about the truth or falsity of the hypothesis. This follows directly from a Bayesian conception of argument or evidence strength. Assuming two mutually exclusive and exhaustive hypotheses:

1. e is an argument for H if $P(e|H) > P(e|\neg H)$
2. the probability of a piece of evidence is determined by total probability (as also found in the denominator of Bayes' rule):
   $$P(e) = P(e|H) * P(H) + P(e|\neg H)P(\neg H)$$
3. now consider the chance of finding an argument a in favour of the hypothesis $a = P(e)$ when assuming an uninformative prior, that is, $P(H) = P(\neg H) = .5$

4. the value of $P(e)$ is determined more by $P(e|H)$ than by $P(e|\neg H)$ as the former is larger than the latter by definition (of what it means to be an argument *for*), given the equal weighting imposed by $P(H) = P(\neg H) = .5$
5. this, in turn, implies that increasing $P(H)$ can only increase $P(e)$, decreasing $P(H)$ can only decrease it, for all $P(e|H) > P(e|\neg H)$

Therefore, as our belief in a hypothesis increases, we –rationally– increasingly expect to find arguments *in favour* of the hypothesis. The opposite is true for arbitrary arguments against.[1]

3.1.2 Strong Arguments are Less Likely than Weak Arguments

The same line of reasoning not only constrains our expectation concerning arguments *for* and *against*, it also constrains our expectations concerning the strength of any such evidence. Specifically, weak evidence is more probable than strong evidence, all other things equal.

1. Again: $P(e) = P(e|H) * P(H) + P(e|\neg H)P(\neg H)$
2. the "strength" of an argument (or piece of evidence) *for* the hypothesis is determined by the difference between $P(e|H)$ and $P(e|\neg H)$ (with bound $P(e|H) = P(e|\neg H)$ in which case the LHR=1, that is, the argument is completely non-diagnostic and receiving it will leave our beliefs unchanged).
3. Assume now an arbitrary value for $P(e|H)$, then increasing $P(e|\neg H)$, thus making the argument less diagnostic necessarily increases $P(e)$, for any uncertain degree of belief in H, that is, $0 < P(H) < 1$:
   (a) Given that $P(\neg H) = 1 - P(H)$, 1. can be rewritten as $P(e) = [P(e|H) - P(e|\neg H)] * P(H) + P(e|\neg H) = P(e|H) * P(H) - P(e|\neg H) * P(H) + P(e|\neg H)$.
   (b) For $0 < P(H) < 1$, however, $P(e|\neg H) > P(e|\neg H) * P(H)$.

[1] In fact, $P(\neg e)$, if observed, would provide an argument *against* H. Because $P(not a) = 1 - P(a)$, we also think counter argument $\neg e$ is less and less likely, as our belief in H increases, as we will see in a moment.

In short, we expect to find both more evidence in favour than against our currently favoured hypothesis, and we expect to find more weak evidence than stronger. For example, if we suspect someone has a particular disease, we expect to find more evidence supporting that conclusion than evidence undermining; and we expect there to be comparatively fewer very strong diagnostic cues than weaker ones. The implications of this for information search will be discussed in the final section below. Before that we consider a second, closely related, issue. These constraints just outlined are about evidence *in expectation*, and how they change given a current degree of belief in H. In other words, they are constraints that follow from our subjective beliefs about the world. We next consider these same issues under an objective Bayesian perspective that looks at how evidence might be generated by the world.

3.2 World Generated Evidence

The preceding section described our expectations about evidence given uncertainty about the truth or falsity of the hypothesis. In actual fact, however, our hypothesis will be either true or false. So we need to consider the implications of this for the distribution of evidence.

Conceptually, this equates to thinking about the world as a stochastic generator of evidence, where the probability with which evidence is generated reflects the world's true causal structure. This notion is, in fact, implemented in a recent Bayesian agent-based model of argument, called NormAN, which models the exchange of arguments over a ground truth world (Assaad et al., 2023). In NormAN, that ground truth world is defined by a Bayesian Belief Network, such as the network seen in Fig. 2.

This well-known network from the literature represents the relations between a number of variables related to whether or not a patient might have lung cancer (node L), such as whether or not the patient is a smoker, has a shortness of breath (dyspnoea), or a shadow on their x-ray. Assaad et al. use this network to generate a world

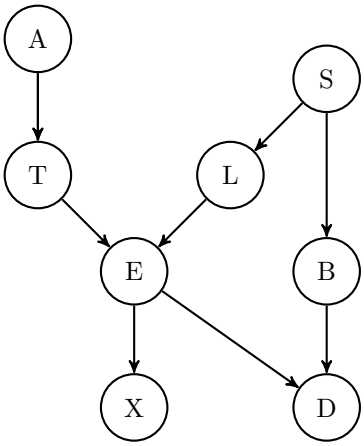


Fig. 2 The 'world model' defined by the so-called 'Asia network'. In this Bayesian causal network, the variables represent whether the patient recently visited Asia (A); has tuberculosis (T), lung cancer (L), either (E), dyspnoea (D), or bronchitis (B); shows problematic chest X-ray (X) results; or smokes (S). All arrows represent positive probabilistic relevance. (Lauritzen & Spiegelhalter, 1988)

state and a set of in principle available evidence (which agents can then obtain and communicate) on a given model run. To this end, they stochastically initialise the hypothesis node to be true or false (in line with its prior probability). They then recompute the marginal probabilities of the evidence nodes in light of that underlying state of the world, and then determine the states of those nodes stochastically in keeping with these new marginal probabilities.

An initialised 'ground truth' world in NormAN is thus a particular instantiation of a causal BN where the combinations of variable states that hold in that particular world follows the probabilistic relationships defined by the graph. In other words, one causal Bayesian Network gives rise to many possible (instantiated) 'worlds'. Moreover, while the Bayesian Network defines the inferential relationships (e.g., that smoking raises the probability of lung cancer), it is the actual state of the evidence node that determines whether it represents and argument for or against. Given that smoking increases the probability of lung cancer, observing that a patient does, in fact, smoke is probability raising for the hypothesis 'lung cancer'; but observing that the patient does not, in fact, smoke lowers that probability.

In short, NormAN embodies in a straightforward way the notion that evidence is generated by the causal structure of the world in a setting where the likelihoods assigned to evidence reflect objective probabilities.

We next work through the implications of this for plausible distributions of evidence. By convention, we will again assume e to represent evidence for and $\neg e$ evidence against H which we assume to be true (if H is false, all points symmetrically flip).

With that in hand, we show that once P(H) = 1, a counter-argument (against H) has to be either (comparatively) weak or (comparatively) improbable. This emerges from the constraints between the various probabilities determining argument strength for binary variables and hypotheses, as set out in Table 2

Strong Argument *for*		Strong Argument *against*	
$P(e\|H)$	$+$	$P(\neg e\|H)$	$= 1$
$>>$		$<<$	
$P(e\|\neg H)$	$+$	$P(\neg e\|\neg H)$	$= 1$

Table 2 Constraint relations between the likelihoods determining the strength of arguments for and against.

Given $P(\neg H) = 1 - P(H) = 0$,

1. There can be no false positives,

   $P(e) = P(e|H) * P(H) + P(e|\neg H)P(\neg H)$ reduces to $P(e) = P(e|H) * P(H) = P(e|H)$

2. There can be no correct rejections, there can only be incorrect rejections

   $P(\neg e) = P(\neg e|H) * P(H) + P(\neg e|\neg H)P(\neg H)$, reduces to $P(\neg e) = P(\neg e|H) * P(H) = P(\neg e|H)$

3. "strong counter-arguments" *in favour* of $\neg H$ require

   $P(\neg e|\neg H) >> P(\neg e|H)$,

   so for there to be such an argument $P(\neg e|H)$ has to be small

4. but in this case its probability of occurrence is constrained by the row relationships in Table 2. The stronger we make the argument by decreasing $P(\neg e|H)$ the more we have to increase the value of $P(e|H)$. That, however, will increase $P(e) = P(e|H) * P(H)$, making $P(\neg e) = 1 - P(e)$ smaller in turn.
5. so either $\neg e$ is comparatively weak or comparatively improbable
6. By contrast: once $P(H) = 1$ there can be arbitrarily strong positive arguments, and they can be arbitrarily probable

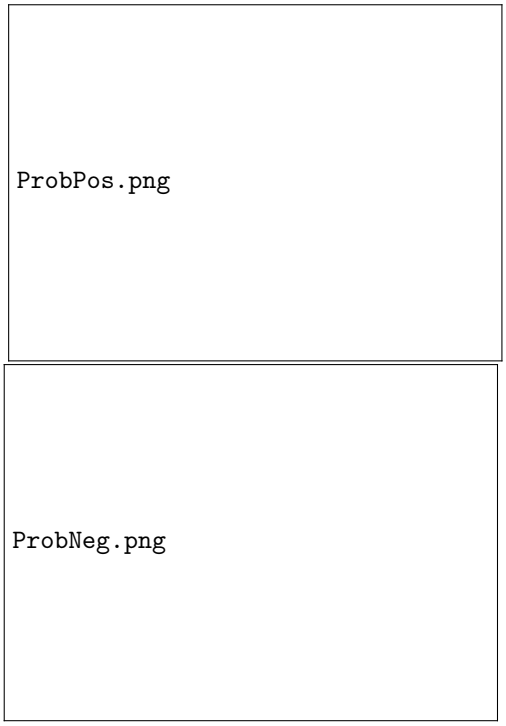


Fig. 3 Two different views on argument strength space as a function of sensitivity and specificity. Argument strength is measured by the LHR, so higher numbers represent stronger arguments. Positive arguments are white, negative arguments are blue. For orientation purposes, the marginal probability of the arguments is also rendered, once in green: marginal probability positive argument, once in red: Marginal probability negative argument. Range for sensitivity and specificity, p = .05 to .95.

Fig 3 plots argument strength (as measured by the likelihood ratio)[2] for both positive arguments *for* and negative arguments *against* over the same range of values for sensitivity and specificity. As can be seen, where a positive outcome is strong, the counter-argument is weaker and vice versa (i.e., one surface lies above the other), bar the equal strength diagonal where both observing e and $\neg e$ give rise to arguments of equal strength (i.e., the two surfaces intersect). Additionally, the plots show the marginal probability of the argument occurring (given that $\mathrm{P}(\mathrm{H} = 1)$, once for positive arguments (green) and once for negative arguments (red). This allows visualisation of the just stated relationship whereby strong counter-arguments are improbable, and, conversely, probable counter-arguments are weak.

Furthermore, for both positive and negative arguments, the relationship described in section 3.1.2 above is recovered: At a given level of probability, there are proportionally more weak arguments than strong arguments.

4 Consequences of Non-Uniform Evidence Distribution

As we just saw, plausible evidence distributions 'bend toward truth'. A Bayesian agent has strong expectations about the likely distribution of evidence, and these follow directly from a simple conception of diagnosticity or argument strength in terms of the likelihood ratio. This has multiple immediate implications for thought about bias and irrationality.

First, it turns the common conception of epistemic 'even-handedness' on its head. The intuition underlying the charge of irrationality levelled at 'hypothesis determined information seeking' is one of even-handedness that is fundamentally at odds both

[2] From the perspective of the same focal hypothesis, H, arguments for and against correspond to LHRs greater and smaller than 1, respectively. So the plot shows the inverse of the latter to have both on the same scale.

with our expectations concerning a world whose causal structure generates evidence, and at odds with expectations flowing from our current subjective beliefs.

Demanding that a cognitive agent strongly convinced of a hypothesis go out and systematically search for disconfirming evidence, is, in effect, sending that agent on what is subjectively a fools errand. It is hard to see how under any rational allocation of resources it would be desirable for a resource-bounded agent to wake up in the morning and think "I've become a bit complacent in my beliefs about the shape of the Earth. In order to attain superior rationality, I will spend my day seeking robust evidence in favour of flat Earth."

Yet that is, in effect, the demand placed by epistemic even-handedness, or, even worse, a focus on falsification. A mistaken demand for even-handedness is embodied in experimental paradigms that chastise participants for search strategies that are sensitive to their current beliefs. For example, (Edwards & Smith, 1996) probed whether participants gave more scrutiny to arguments that conflicted with their current beliefs, as opposed to arguments that concurred. Edwards and Smith considered it to be indicative of bias that participants engaged in a more thorough search for their own (counter)arguments to an experimenter-provided argument when it was incompatible with their beliefs than when the experimenter's argument was compatible, as evidenced by longer deliberation time. Likewise, they considered it to be evidence of bias that among the actual arguments participants generated, more were refutational (rather than supportive) in nature when the experimenter argument for scrutiny was incompatible with prior beliefs than when it is compatible.

A mistaken demand for even-handedness also underlies intuitive, but problematic, intuitions about how evidence exchange should affect deliberating groups. Phenomena such as group shifts to extremity have attracted long-standing interest precisely because it seemed puzzling that group members exchanging arguments would shift beyond an initial group mean (Broncano-Berrocal & Carter, 2021; Myers & Lamm,

1976). However, evidence agents receive from the world will not be 'unbiased' in the sense of equally distributed for and against a target claim, nor will the distribution of evidence in a group of rational agents be unrelated to their current beliefs (Assaad et al., 2023; Hahn, 2024), nor will their communications be 'unbiased'.

Our very notions of cooperative communication as underlie natural language pragmatics (Grice, 1969; Leech, 2016) demand that we communicate to others strong evidence in support of what we actually believe in preference to weak evidence. Likewise, communicating counter-evidence without some indication of the fact that it runs counter to what we currently believe most likely to be true would seem misleading.

Counter to the judgment underlying 'myside-bias' (Baron, 1995; Mercier, 2022), we need to factor in that rational agents that are not wholly agnostic on an issue not only will, overall, possess more evidence supporting their current belief than against, but that they *should* communicate their (subjectively) best evidence (see (Assaad et al., 2023; Assaad & Hahn, 2024)). Myside bias, to that effect, is not bias, but helpful communication.[3]

This is important, because the fact that we expect plausible evidence distributions to 'bend toward truth' justifies our belief that the acquisition of evidence and arguments will lead us toward truth, given rational evaluation. This, however, holds only if the way that evidence is dispensed is itself unbiased.

To the extent that we need to search for evidence, this may well not be the case, or become unbiased only in the limit. The fact that our subjective expectations about the distribution of evidence interact with our current degree of belief may give rise to

[3]It should be acknowledged that the Bayesian framework concerns arguments about facts, whereas some empirical demonstrations of myside bias involve opinions. Strictly speaking, some preceding considerations do not directly apply. In particular, there is no reason to believe that the causal structure of the world need deliver arguments that speak in favour a particular valuation at any particular rate relative to those against. That said, it is often assumed that seemingly biased behavioural strategies such as a positive test strategy are generalised heuristics that reflect the fact that the circumstances under which they are adaptive are often met, as opposed to assuming that a fully rational response is calculated on each occasion. From that perspective, what holds for facts seems relevant also to opinions, if for no other reason than the fact that the differences between the two seems subtle to many lay reasoners, see also (Hahn, 2024).

a rachet, whereby resource rational selection of evidence sources may carry the risk of selectively further affirming driving beliefs.

As discussed earlier, a similar rachet exists with respect to source reliability, whereby we might use belief congruence to shape our beliefs about the reliability of a source (Bovens & Hartmann, 2003; Merdes et al., 2021; Olsson, 2011). This, in itself, can lead otherwise unbiased agents to seemingly biased, polarised states, merely by contingent fate of experiential history (Hahn et al., 2024; Madsen et al., 2018; Olsson, 2013; Pallavicini et al., 2021). It also provides one perspective on homophily and affiliative sorting in social networks as a prime cause of biased receipt of evidence. On this view, we simply choose to communicate with those who we believe to be more reliable.

An alternative, not mutually exclusive, possibility for such phenomena also exists, namely that I do not believe those I seek out for belief congruence to be more reliable, but rather that I simply believe them to be more likely to be in possession of strong evidence. The crucial conceptual difference, here, is that this could give rise to affiliative sorting even in circumstances where an agent sees no difference in sources' reliability or does not view sources' reliability through the prism of belief congruence.

In summary, it is the contention of this paper that the existence of constraints on plausible distributions of evidence from the perspective of a rational agent provides a fundamental back drop against which research on bias and human irrationality must proceed. While the role of plausible evidence distributions has crept up repeatedly in the history of research on confirmation bias to date, it is, arguably, still not properly appreciated in as much as still popular benchmarks for unbiasedness, or even-handedness, seem fundamentally at odds with those constraints.

Finally, this has obscured deeper consideration of the ways our own actions, both as information seekers and as communicators, may interact with those constraints to achieve particular information, and with that, belief outcomes.

5 Conclusions

In this paper, it was argued that constraints on the distribution of evidence provide a conceptual theme uniting counter-arguments against charges of confirmation bias that run through multiple different tasks and phenomena. These span information search in: rule induction (Klayman & Ha, 1987; Wason, 1960), the evaluation of conditional hypotheses (Oaksford & Chater, 1994; Wason, 1968)), and the determination of contingencies (Crupi et al., 2009; Doherty, Mynatt, Tweney, & Schiavo, 1979). They also include the inference of biased, confirmatory processing from patterns in estimates of correlation (Chapman & Chapman, 1967; Fiedler, 2000) or seemingly optimistic judgments of personal risk (Harris & Hahn, 2011; Weinstein, 1980), as well as patterns of belief revision in optimistic belief updating (Shah et al., 2016; Sharot et al., 2011).

It was then demonstrated that there are expectations on the plausible distribution of evidence that concern the probability of arguments as a function of polarity (arguments for or against) and argument strength that follow directly from a Bayesian conception of evidence.

Characterising expectations on the plausible distribution of evidence in this way affords insight into aspects of confirmation bias such as 'myside bias' that have seen little challenge relative to topics like hypothesis testing. These expectations also afford insight into group level phenomena that, until recently, have tended to play a more minor role in traditionally individual-level focussed rationality research. In particular, the plausible distribution of evidence both explains the prevalence of shifts to extremity in group deliberation (Assaad et al., 2023; Hahn, 2024) and it paves the way for novel forms of rational group polarisation that rest on the interaction between strategies for selecting what arguments to communicate and the distributional characteristics of the arguments that are likely to be available (Assaad et al., 2023; Assaad & Hahn, 2024).

Together, these considerations highlight the multiple ways in which the widespread attribution of 'confirmation bias' often outruns the specific, empirical support, and they re-emphasise the necessity of considering in detail how a rational agent will fare with task before attributing bias.

6 CoI Statement

On behalf of all authors, the corresponding author states that there is no conflict of interest.

References

Assaad, L., Fuchs, R., Jalalimanesh, A., Phillips, K., Schoeppl, L., Hahn, U. (2023). A bayesian agent-based framework for argument exchange across networks. *arXiv preprint arXiv:2311.09254*, .

Assaad, L., & Hahn, U. (2024). Rational polarization: Sharing only one's best evidence can lead to group polarization. *Proceedings of the annual meeting of the cognitive science society* (Vol. 46).

Baron, J. (1995). Myside bias in thinking about abortion. *Thinking & Reasoning*, *1*(3), 221–235.

Bhatia, J.-S., & Oaksford, M. (2015). Discounting testimony with the argument ad hominem and a bayesian congruent prior model. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *41*(5), 1548.

Bovens, L., & Hartmann, S. (2003). *Bayesian epistemology*. Oxford University Press.

Broncano-Berrocal, F., & Carter, J.A. (2021). *The philosophy of group polarization: Epistemology, metaphysics, psychology.* Routledge.

Chapman, L.J., & Chapman, J.P. (1967). Genesis of popular but erroneous psychodiagnostic observations. *journal of Abnormal Psychology*, *72*(3), 193.

Chater, N., & Oaksford, M. (2000). The rational analysis of mind and behavior. *Synthese*, *122*, 93–131.

Chater, N., Oaksford, M., Hahn, U., Heit, E. (2010). Bayesian models of cognition. *Wiley Interdisciplinary Reviews: Cognitive Science*, *1*(6), 811–823.

Corner, A., Hahn, U., Oaksford, M. (2011). The psychological mechanism of the slippery slope argument. *Journal of Memory and Language*, *64*(2), 133–152.

Crupi, V., Tentori, K., Lombardi, L. (2009). Pseudodiagnosticity revisited. *Psychological Review*, *116*(4), 971.

Doherty, M.E., Mynatt, C.R., Tweney, R.D., Schiavo, M.D. (1979). Pseudodiagnosticity. *Acta psychologica*, *43*(2), 111–121.

Edwards, K., & Smith, E.E. (1996). A disconfirmation bias in the evaluation of arguments. *Journal of personality and social psychology*, *71*(1), 5.

Eva, B., & Hartmann, S. (2018). Bayesian argumentation and the value of logical validity. *Psychological Review*, *125*(5), 806.

Evans, J.S.B., & Over, D.E. (1996). Rationality in the selection task: Epistemic utility versus uncertainty reduction.

Fiedler, K. (2000). Illusory correlations: A simple associative algorithm provides a convergent account of seemingly divergent paradigms. *Review of General Psychology*, *4*(1), 25–58.

Gold, E.M. (1967). Language identification in the limit. *Information and control*, *10*(5), 447–474.

Gopnik, A., & Bonawitz, E. (2015). Bayesian models of child development. *Wiley interdisciplinary reviews: cognitive science*, *6*(2), 75–86.

Green, D.M., Swets, J.A., et al. (1966). *Signal detection theory and psychophysics* (Vol. 1). Wiley New York.

Grice, H.P. (1969). Utterer's meaning and intention. *The philosophical review*, *78*(2), 147–177.

Hahn, U. (2020). Argument quality in real world argumentation. *Trends in Cognitive Sciences*, *24*(5), 363–374.

Hahn, U. (2024). Individuals, collectives, and individuals in collectives: The ineliminable role of dependence. *Perspectives on Psychological Science*, *19*(2), 418–431.

Hahn, U. (2026). Human rationality. *Annual Review of Psychology*, *77*, .

Hahn, U., & Harris, A.J. (2014). What does it mean to be biased: Motivated reasoning and rationality. *Psychology of learning and motivation* (Vol. 61, pp. 41–102). Elsevier.

Hahn, U., Merdes, C., von Sydow, M. (2024). Knowledge through social networks: Accuracy, error, and polarisation. *Plos one*, *19*(1), e0294815.

Hahn, U., & Oaksford, M. (2008). Inference from absence in language and thought. *The probabilistic mind: Prospects for Bayesian cognitive science*, 121–42.

Harris, A.J., & Hahn, U. (2011). Unrealistic optimism about future life events: a cautionary note. *Psychological review*, *118*(1), 135.

Hsu, A.S., Chater, N., Vitányi, P. (2013). Language learning from positive evidence, reconsidered: A simplicity-based approach. *Topics in cognitive science*, *5*(1), 35–55.

Jeffrey, R.C. (1990). *The logic of decision*. University of Chicago press.

Jern, A., Chang, K.-M.K., Kemp, C. (2014). Belief polarization is not always irrational. *Psychological review*, *121*(2), 206.

Klayman, J., & Ha, Y.-W. (1987). Confirmation, disconfirmation, and information in hypothesis testing. *Psychological review*, *94*(2), 211.

Laming, D. (1996). On the analysis of irrational data selection: A critique of oaksford and chater (1994).

Lauritzen, S.L., & Spiegelhalter, D.J. (1988). Local computation with probabilities on graphical structures and their application to expert systems (with discussion). *Journal of the Royal Statistical Society. Series B (Methodological)*, *50*(2), 157–224. Retrieved from https://www.jstor.org/stable/2345762

Leech, G.N. (2016). *Principles of pragmatics*. Routledge.

Lench, H.C., & Ditto, P.H. (2008). Automatic optimism: Biased use of base rate information for positive and negative events. *Journal of Experimental Social Psychology*, *44*(3), 631–639.

Lord, C.G., Ross, L., Lepper, M.R. (1979). Biased assimilation and attitude polarization: The effects of prior theories on subsequently considered evidence. *Journal of personality and social psychology*, *37*(11), 2098.

Madsen, J.K., Bailey, R.M., Pilditch, T.D. (2018). Large networks of rational agents form persistent echo chambers. *Scientific reports*, *8*(1), 12391.

Mercier, H. (2022). Confirmation bias–myside bias. *Cognitive illusions* (pp. 78–91). Routledge.

Merdes, C., Von Sydow, M., Hahn, U. (2021). Formal models of source reliability. *Synthese*, *198*, 5773–5801.

Myers, D.G., & Lamm, H. (1976). The group polarization phenomenon. *Psychological bulletin*, *83*(4), 602.

Nickerson, R.S. (1998). Confirmation bias: A ubiquitous phenomenon in many guises. *Review of general psychology*, *2*(2), 175–220.

Oaksford, M., & Chater, N. (1994). A rational analysis of the selection task as optimal data selection. *Psychological review*, *101*(4), 608.

Oaksford, M., & Chater, N. (1996). Rational explanation of the selection task.

Olsson, E.J. (2011). A simulation approach to veritistic social epistemology. *Episteme*, *8*(2), 127–143.

Olsson, E.J. (2013). A bayesian simulation model of group deliberation and polarization. *Bayesian argumentation* (pp. 113–133). Springer.

Pallavicini, J., Hallsson, B., Kappel, K. (2021). Polarization in groups of bayesian agents. *Synthese*, *198*, 1–55.

Perfors, A., Tenenbaum, J.B., Wonnacott, E. (2010). Variability, negative evidence, and the acquisition of verb argument constructions. *Journal of child language*, *37*(3), 607–642.

Pettigrew, R. (2016). *Accuracy and the laws of credence*. Oxford University Press.

Shah, P., Harris, A.J., Bird, G., Catmur, C., Hahn, U. (2016). A pessimistic view of optimistic belief updating. *Cognitive Psychology*, *90*, 71–127.

Sharot, T., Korn, C.W., Dolan, R.J. (2011). How unrealistic optimism is maintained in the face of reality. *Nature neuroscience*, *14*(11), 1475–1479.

Sunstein, C.R. (2009). *Going to extremes: How like minds unite and divide*. Oxford University Press.

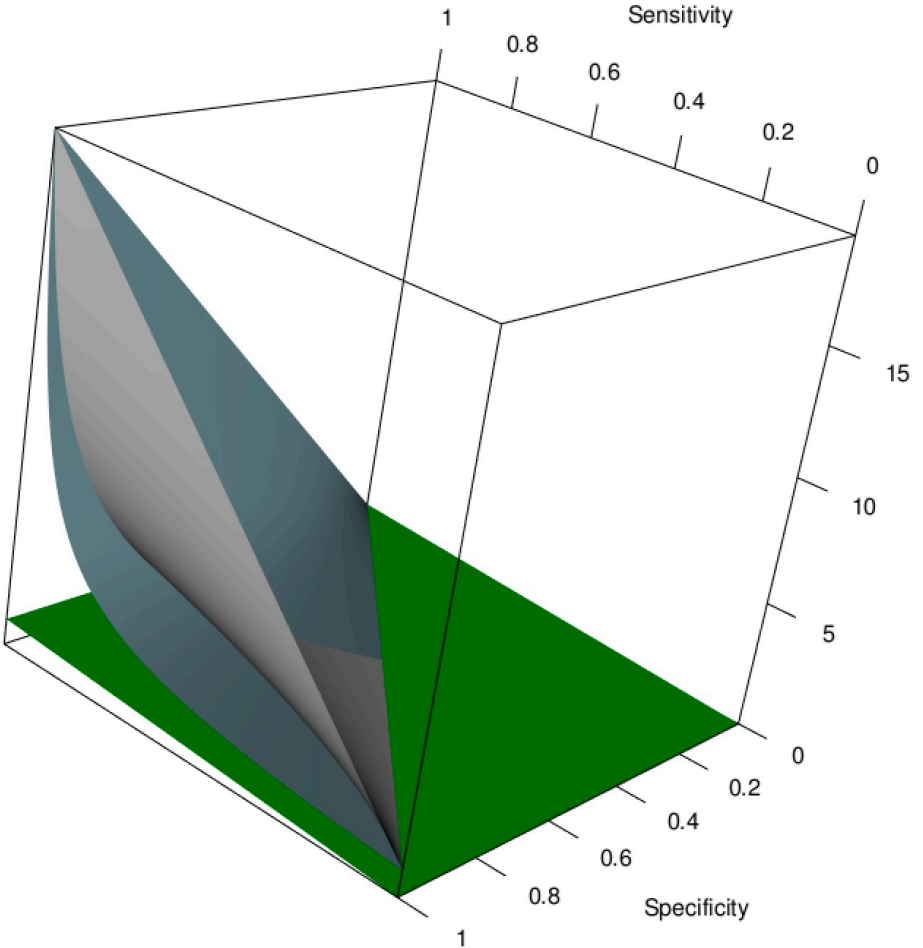
Wason, P.C. (1960). On the failure to eliminate hypotheses in a conceptual task. *Quarterly journal of experimental psychology*, *12*(3), 129–140.

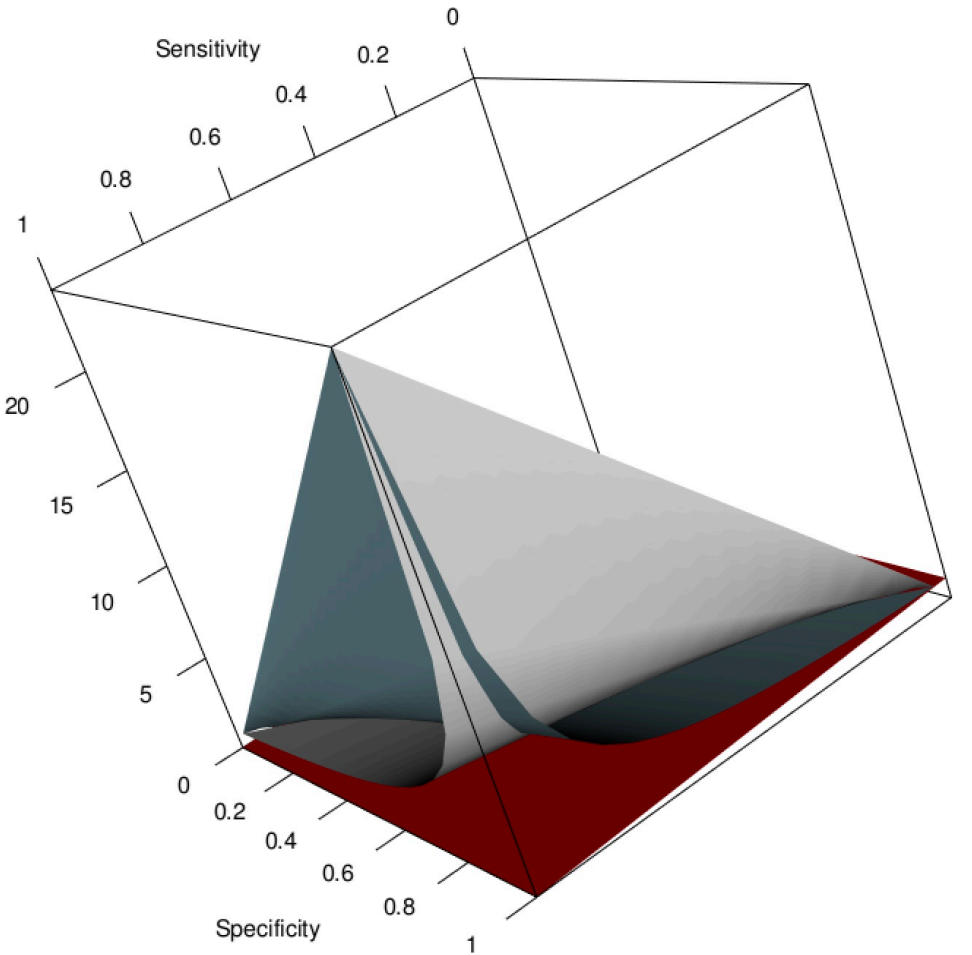
Wason, P.C. (1968). Reasoning about a rule. *Quarterly journal of experimental psychology*, *20*(3), 273–281.

Weinstein, N.D. (1980). Unrealistic optimism about future life events. *Journal of personality and social psychology*, *39*(5), 806.

Williamson, J. (2010). *In defence of objective bayesianism*. OUP Oxford.

Phenomenon		Example	Counter-evidence
1. Hypothesis-Determined Information Seeking & Interpretation			
	1.1 Restriction of attention to a favored hypothesis.	consider only *P(D |H)*, not *P(D |notH)*, e.g.; Doherty, Mynatt, Tweney, & Schiavo (1979) - pseudodiagnosticity-bias	Crupi, Tentori & Lombardi (2009)
	1.2 Preferential treatment of evidence supporting existing beliefs.	my-side bias: tendency to produce reasons for favored side e.g., Baron (1995)	Mercier (2022)
	1.3 Looking only or primarily for positive cases.	Tendency to ask questions for which answer would be 'yes' if hypothesis is true: Wason (1960)	Klayman & Ha (1987)
	1.4 Overweighting positive confirmatory instances.	e.g., Gilovich (1983)	
	1.5 Seeing what one is looking for.	e.g., effects of expectations on social perception Kelley (1950)	Lenski & Leggett (1960) question reponse tendency in acquiescense to interrogator hypothesis.
	1.6 Remembering what one expects	Eagly et al. (1999)	Hart et al. (2009)
	1.7 Illusory correlation	Chapman & Chapman (1967)	Fiedler (2000)
2. Wason selection task and formal reasoning		Failure to pursue falsificationist strategy for conditional reasoning, Wason (1968)	Oaksford & Chater (1994)
3. The Primacy Effect and Belief Persistence		Resistance of belief/opinion to change e.g., Pitz, Downing & Reinhold's (1967) inertia effect; Lord, Ross & Lepper (1979) 'biased assimilation'	Kemp et al. (2014); Hahn & Harris (2014)
4. Overconfidence and the illusion of validity		e.g., Lichtenstein & Fischhoff (1977)	Erev, et al. (1994)


Sensitivity
1
0.8
0.6
0.4
0.2
0
15
10
5
0
0.2
0.4
0.6
0.8
1
Specificity


Sensitivity
0
0.2
0.4
0.6
0.8
1
20
15
10
5
0
0.2
0.4
0.6
0.8
1
Specificity